

Basic Review of Vec. Spaces:

$$V = (\mathbb{R}^2, +, \cdot)$$

$$(x, y), (x', y') \in \mathbb{R}^2$$

$$\langle (x, y), (x', y') \rangle_{\mathbb{R}^2} = xx' + yy'$$

$$\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\|(x, y)\|^2 \equiv x^2 + y^2$$

* Basis elements $\{e_1, e_2\}$

* Metric: $g : V \times V \rightarrow \mathbb{R}$

Non-degeneracy: Means, for $g : V \times V \rightarrow \mathbb{R}$

$g(u, \cdot) : V \rightarrow \mathbb{R}$ is non zero for any $u \in V$

* Orthogonal Complement to U , $U \subseteq V$

$$U^\perp := \{v \in V \mid g(v, u) = 0\}$$

$$U \oplus U^\perp = V$$

Linear Symplectic Structures:

* Skew-symmetric, non degenerate bilinear map

$$\omega : V \times V \rightarrow \mathbb{R} \quad \leftarrow \text{field} = (\mathbb{R} \text{ or } \mathbb{C})$$

ex: ω on $\mathbb{R}^2$, $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$

$$\omega((x, y), (x', y')) \equiv xy' - yx'$$

which we can write as

$$\omega((x, y), (x', y')) \mapsto \begin{vmatrix} x & y \\ x' & y' \end{vmatrix} = \det \begin{pmatrix} x & y \\ x' & y' \end{pmatrix}$$

* In general, $\mathbb{R}^{2n}$ has a symplectic form:

$$\omega: \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$$

Defined by

$$\omega_1((x_1, y_1, \dots, x_n, y_n), (x'_1, y'_1, \dots, x'_n, y'_n)) \\ = \sum_{i=1}^n (x_i y'_i - y_i x'_i)$$

* In Matrix form:

$$\omega_n(\vec{a}, \vec{b}) = \bigoplus_{i=1}^n \omega_1((x_i, y_i), (x'_i, y'_i))$$

where $\vec{a} = (x_1, y_1, \dots, x_n, y_n) \in \mathbb{R}^{2n}$

$\vec{b} = (x'_1, y'_1, \dots, x'_n, y'_n) \in \mathbb{R}^{2n}$

(i) ω_1 and hence, ω_n is skew symm.

proof: Let $u, v \in \mathbb{R}^2$ s.t. $u \neq v$

then $u = (u_1, u_2)$

$v = (v_1, v_2)$

Matrix form

$$\omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\omega(u, v) = \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} = u_1 v_2 - u_2 v_1 \quad X^T \omega X \in \mathbb{R}$$

$$= -(v_1 u_2 - v_2 u_1) = - \begin{vmatrix} v_1 & v_2 \\ u_1 & u_2 \end{vmatrix}$$

$$= -\omega(v, u)$$

$$\therefore \omega(u, v) = -\omega(v, u)$$

(ii) If $u = v$, then

$$\omega(u, u) = -\omega(u, u) \Leftrightarrow \omega(u, u) = 0$$

Def: Symplectic Condition of a Scheme

$U \subseteq V$ is defined as

$$U^\omega \equiv \{v \in V \mid \omega(v, u) = 0\}$$

* Define $\langle u \rangle \equiv \langle u, u \rangle = \omega(u, u)^2$

$$\langle u \rangle^\omega \equiv \omega(u, u) = 0$$

• $\langle u \rangle^\omega$ cannot be all of V

$$\langle u \rangle + \langle u \rangle^\omega \neq V$$

Lemma: For any subspace $U \subseteq V$ of a symplectic space (V, ω)

(i) $\dim(U) + \dim(U^\omega) = \dim(V)$

(ii) $U \oplus U^\omega = V$ iff $\omega|_U$ is a linear symplectic form.
In this case U is called a symplectic subspace.

(iii) $(U^\omega)^\omega = U$. U is symplectic iff U^ω is symplectic.

PROOF: We have $\omega: V \times V \rightarrow K$

We have the isomorphism formed by

$$\omega: V \cong V^*$$

(a) i.e. for any vector $x \in V$ we have an element of V^* formed via

$$x \mapsto \omega(x, \cdot) \in V^*$$

(b) Likewise for any element $\alpha \in V^*$, $\alpha: V \rightarrow K$ for which $x \mapsto \alpha(x) \in K$

Thus $\omega: V \rightarrow V^*$. This is exactly the same for the metric in CR.

If we have a "vector" V^a we can obtain a "covector" via "lowering" of this index:

$$V \ni V^a \mapsto g_{ab} V^a \equiv \omega_b \in V^*$$

for which $g: V \rightarrow V^*$

→ Now, take U^ω , then
 $\omega: U^\omega \rightarrow (U^\omega)^*$

$$\omega(\tilde{u}) = \omega(\tilde{u}, \cdot) \equiv \alpha$$

Now, since $\tilde{u} \in U^\omega$ then for any $u \in U$ we must have

$$\omega(\tilde{u}, u) = 0$$

or

$$\alpha(u) = 0$$

We conclude the image of $\omega(U^\omega)$ to be

$$\omega(U^\omega) = \text{Ann}(U) \equiv \{\alpha \in V^* \mid \alpha(U) = 0\}$$

Note $\text{Ann}(U) = \text{Kern}(\hat{r})$, where

$\hat{r}$ is the restriction map

$$\hat{r}: V^* \rightarrow U^*$$

$$\hat{r}(\alpha) = \alpha|_U.$$

* We can make the composition:

$$V \rightarrow V^* \rightarrow U^*$$

$$(\hat{\rho}\omega): V \rightarrow U^*$$

is surjective.

Proof? Suppose $\exists x \in V$ s.t.

$$(\hat{\rho}\omega)(x) = 0$$

$$\text{then } \hat{\rho}(\omega(x)) = \hat{\rho}(\omega(x, \cdot))$$

$$\omega(x) \equiv \alpha_x \in U^*$$

$$\text{thus } \hat{\rho}(\alpha_x) = (\alpha_x)|_U : U \rightarrow K$$

Because $U \subset V$ we can always find an element of V s.t. we have a map in U^* .

$$\text{C) } \ker(\hat{\rho}\omega) = U^\omega$$

$$\text{Recall, } \ker(\hat{\rho}\omega) \equiv \{v \in V \mid (\hat{\rho}\omega)(v) = 0\}$$

Proof: ($\Rightarrow$)

Suppose $x \in \ker(\hat{\rho}\omega)$ then we must show $x \in U^\omega$.

Now, if $x \in \ker(\hat{\rho}\omega)$ then

$$(\hat{\rho}\omega)(x) = 0$$

$$\hat{\rho}(\omega(x, \cdot)) = \omega(x, \cdot)|_U = 0$$

Meaning if we take any $u \in U$ that $\omega(x, u) = 0 \quad \forall u \in U$

$$\therefore x \in U^\omega$$

($\Leftarrow$) Suppose $x \in U^\omega$, then

$$(\hat{\rho}\omega)(x) = \omega(x, \cdot)|_U$$

$$\omega(x, u) = 0 \quad \forall u \in U \text{ by def.}$$

thus, $x \in \ker(\hat{\rho}\omega)$. $\square$

Putting all of this together we find

$$\dim(V) = \dim(U^\omega) + \dim(U^*)$$

Since for a morphism

$$(\tilde{\rho}\omega): V \rightarrow U^*$$

$$V/\ker(\tilde{\rho}\omega) \cong U^*$$

$$\Rightarrow \dim(V/\ker(\tilde{\rho}\omega)) = \dim(U^*)$$

$$\dim(V) - \dim(\ker(\tilde{\rho}\omega)) = \dim(U^*)$$

$$\therefore \dim = \dim(U^\omega) + \dim(U^*)$$

where we found $\ker(\tilde{\rho}\omega) = U^\omega$

and $|U^*| < \infty \Leftrightarrow \dim(U^*) = \dim(U) \square$

* $U \oplus U^\omega = V$ is equivalent to $U \cap U^\omega = \{0\}$

The latter is equivalent to

$$\omega(u, v) \neq 0 \quad \forall u \in U$$

since $0 \in U^\omega$ and so $\omega|_U$ is non-deg.

* Finally, $\omega(u, U^\omega) = 0 \quad \forall u \in U$ we have

$$U \subseteq (U^\omega)^\omega$$

$$\dim(U) = \dim(U) + \dim(U^\omega) = \dim(U^\omega) + \dim((U^\omega)^\omega)$$

* With this decomposition Lemma we may find a normal form for a symplectic structure

(i) Take a non-zero vector $u \in V$. Since

$$\omega(u, \cdot): V \rightarrow K \neq 0 \quad \exists v_1 \in V \text{ s.t.}$$

$$\omega(u, v_1) = 1$$

The subspace

$$U_1 \cong \langle u, v_1 \rangle$$

is symplectic?

Q: Is this notation really supposed to be $U_1 = \text{span}\{u, v\}$? I ask myself as $\langle 0, 0 \rangle \in K$ is a field element i.e. a number

* The decomposition ∇ found

above:

$$\omega_n(\vec{a}, \vec{b}) = \bigoplus_{i=1}^n \omega_i((x_i, y_i), (x'_i, y'_i))$$

is exactly the decomp on pgs 9/5

where

$$\omega_i = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ as we found}$$

and

$$\omega_n = \bigoplus_{i=1}^n \omega_i = \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} & & 0 \\ & \ddots & \\ 0 & & \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{pmatrix}$$

* A Vector Space V admitting a linear symplectic structure must be even dimensional.

Linear Presymplectic & Poisson structures

Def: A linear Presymplectic structure on V is a pair (V, ω) where ω is possibly degenerate and $\omega: V \times V \rightarrow K$ is skew-symmetric.

→ Normal form of a pre symplectic structure

$$w: V \rightarrow V^*$$

(ii) Consider $\text{Ker}(w)$ & find a complement W
 $W \equiv \text{Ker}(w)^\perp, W \subset V$

$w|_W$ is a symplectic form

Proof: $\text{Ker}(w) \equiv \{ \alpha \in V^* \mid \alpha(z) = 0, z \in V \}$

$$w(x, \cdot) \equiv \alpha_x \dots$$

Same as above...

→ On W we can find a basis s.t.

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

(iii) Add this basis $\text{Span}\{w_1, \dots, w_d\} = W$
 to any basis of $\text{Ker}(w)$ to obtain
 a full basis of V . For fact
 $w(\text{Ker}(w), W) = 0$

and so

$$w_{\text{pre}} = \begin{pmatrix} 0 & & \\ & 0 & 1 \\ & -1 & 0 \end{pmatrix}$$

(iv) Given w , we have an isomorphism
 $V \rightarrow V^*$

• Invert this iso s.t.

$$w^{-1}: V^* \rightarrow V$$

We can regard w^{-1} as a map

$$w^{-1} \equiv: \pi: V^* \times V^* \rightarrow K$$

ASIDE: (Poisson vs. Symplectic)

$$\pi: V^* \times V^* \rightarrow K \quad \text{vs.} \quad w: V \rightarrow V^*$$

→ We have the so called "musical isomorphism"

$$\pi^\# : V^* \rightarrow V$$

$$\pi^\#(\alpha) \mapsto i_\alpha \pi$$

$$i : V^* \times \Omega^p(M) \rightarrow \Omega^{p-1}(M) \quad [\text{Interior product}]$$

$$i(\alpha, \pi) \equiv i_\alpha \pi \equiv \pi(\alpha, \underbrace{\cdot, \dots, \cdot}_{p-1})$$

* $\pi^\#$ is nondegenerate if it is an isomorphism

$\omega : V \times V \rightarrow K$, determines the map

$$\omega^\flat : V \rightarrow V^*$$

$$v \mapsto i_v \omega$$

Lemma: There is a 1:1 correspondence between non-deg. bivectors π and non-deg 2-forms $\omega \in \Omega^2(M)$:

$$\omega^\flat = (\pi^\#)^{-1} \iff \pi^\# = (\omega^\flat)^{-1}$$

where

$$[\pi, \pi](\alpha, \beta, \gamma) = -2d\omega(\pi^\#(\alpha), \pi^\#(\beta), \pi^\#(\gamma))$$

$$\alpha, \beta, \gamma \in V^*$$

→ I'm personally not super interested in proving this. Mostly wanted to know the exact form of the relationship between these two structures.

* Interesting side notes:

(c) $[\cdot, \cdot]$ is defined as the Schouten bracket

$[X, Y] = X \circ Y - (Y \circ X)$
 $X \in \mathcal{X}^1(M)$ } multivector fields
 $Y \in \mathcal{X}^1(M)$ } (i.e. tensors)

(ii) $f, g \in C^\infty(M)$
 $\{f, g\} = \sum_{i,j=1}^n \pi^{ij}(x) \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^j}$
 (Poisson structure)

(iii) Cool Relationship
 $0 = \frac{1}{2} [\pi, \pi] (df_1, df_2, df_3) = \text{Jacobi}(f_1, f_2, f_3)$

$\text{Jacobi}(f_1, f_2, f_3) \stackrel{!}{=} [\pi, \pi] = 0$

Linear $\mathbb{C}$ -structures:

* A linear $\mathbb{C}$ -structure on a real vec. space
 V is an endomorphism $J \in \text{End}(V)$ s.t.
 $J^2 = -\text{id}_V$

(Example: $\mathbb{R}^2 \cong \mathbb{C}$, note $V = \mathbb{R}^2$

$J: V \rightarrow V$

$J(a, b) = (-b, a)$

in local coords, $x = a + ib \in \mathbb{C}$
 then (a, b)

$i x = -b + i a : (-b, a)$

$\rightarrow J$ is eq. to acting on an element
 of $\mathbb{C}$ w/ i .

* $U \subset V$ then if $J(U) \subseteq U$ and since J is an endomorphism $J(U) = U$. $J|_U$

proof: Let $\dim(U) = 2n$, then

$$\perp (a_1, \dots, a_{2n}) \mapsto J(a_1, \dots, a_{2n}) \in U \dots$$

• Let $v \in V$, then add $Jv \in V$

$$Jv = \lambda v \iff J(Jv) = \lambda^2 v$$

$$-v = \lambda^2 v$$

Choose $w \in V \setminus \langle v, Jv \rangle$

Jw is linearly independent from v, Jv , and $w, \dots$

* We obtain the natural basis:

$$(v_1, Jv_1, \dots, v_m, Jv_m)$$

$$\rightarrow (Jv_1, \dots, Jv_m, v_1, \dots, v_m)$$

$$J \begin{pmatrix} v_1 \\ \vdots \\ v_m \\ Jv_m \end{pmatrix} = \begin{pmatrix} -v_1 \\ \vdots \\ -Jv_m \\ 0 \end{pmatrix}$$

$$J = \begin{pmatrix} 0 & \mathbb{1}_{m \times m} \\ -\mathbb{1}_{m \times m} & 0 \end{pmatrix}$$

Complexification: $V^{\mathbb{C}} = V \otimes V \cong \{u + iv \mid u, v \in V\}$

Scalar product

$$\langle \cdot, \cdot \rangle_{\mathbb{C}} : V^{\mathbb{C}} \rightarrow \mathbb{C}$$

$$\langle a + ib, u + iv \rangle = au - bv + i(av + bu)$$

$$\dim_{\mathbb{R}}(V) = \dim_{\mathbb{C}} V^{\mathbb{C}}$$

* $J \in \text{End}(V)$ extends to an endomorphism

on $V \in V \otimes \mathbb{C}$

$$u + iv \mapsto Ju + iJv$$

$$J^2 + \mathbb{I} = 0$$

∴ For eigen vectors

$$(J - i\mathbb{I})(J + i\mathbb{I}) = 0$$

We can diagonalized J to $\pm i$ eigenspaces

• Let $V \in V^{(1,0)}$ i.e.

$$JV = iV$$

if $V = a + ib$, then

$$JV = iV = -b + ia$$

This implies

$$J\bar{V} = -i\bar{V} \Leftrightarrow$$

$$J(u - iv) = \dots$$

$$V^{1,0} \equiv \{v - iJv \mid v \in V\}$$

$$V^{0,1} \equiv \{v + iJv \mid v \in V\}$$

$$\dim_{\mathbb{R}} V^{1,0} = \dim_{\mathbb{R}} V$$

$$\dim_{\mathbb{C}} V^{1,0} = \dim_{\mathbb{C}} V^{0,1} = \frac{1}{2} \dim_{\mathbb{C}} V^{\mathbb{C}}$$

$$V = V^{1,0} \oplus V^{0,1}$$

We have the isomorphisms:

$$(V, J) \cong (V^{1,0}, J|_{V^{1,0}})$$

$$(V, -J) \cong (V^{0,1}, J|_{V^{0,1}})$$

of real vect. spaces w/ linear $\mathbb{C}$ -str.

$$\varphi: V \rightarrow V^{1,0}$$

$$\varphi(v) := v - iJv$$

$$V \xrightarrow{\varphi} V^{1,0}$$

$$\begin{array}{ccc} J \downarrow & & \downarrow J|_{V^{1,0}} \\ V & \xrightarrow{\varphi} & V^{1,0} \end{array}$$

$$\left\{ \begin{array}{l} V^{1,0} \cap \overline{V^{1,0}} = \emptyset \end{array} \right.$$

Lemma 1.4: A subspace $L \subset V^{\mathbb{C}}$ s.t.

$\dim_{\mathbb{R}} L = \dim_{\mathbb{R}} V$, $L \cap \bar{L} = \{0\}$ determines a decomposition

$$V_{\mathbb{C}} = L \oplus \bar{L}$$

and the subspace $V \subset V_{\mathbb{C}}$ corresponds to the set $\{l + \bar{l} \mid l \in L\}$.

Proof: $L + \bar{L} \subseteq V_{\mathbb{C}}$ --

Proposition 1.5: $L \subset V_{\mathbb{C}}$ s.t. $\dim_{\mathbb{R}} L = \dim_{\mathbb{R}} V$ and $L \cap \bar{L} = \{0\}$, $\exists \mathcal{T}$ s.t. $L = V^{+0}$

Proof: Define $\mathcal{T}$ of $V = \{l + \bar{l} \mid l \in L\}$

$$\text{as } l + \bar{l} \mapsto il - i\bar{l}$$

$$\mathcal{T}(l + \bar{l}) \equiv il - i\bar{l}$$

then $\mathcal{T}: V \rightarrow V$.

Tensors & Their Alg's:

- $\text{Span}\{e_i\} = V$, $B: V \times V \rightarrow \mathbb{R}$

$$B(e_i, e_j)$$

Note $\{(e_i, e_j)\}$ is not $V \times V$ since there are n^2 elements in $\{(e_i, e_j)\}$ but

$\dim(V \times V) = 2n$ given by the basis $\{(e_1, 0), (0, e_1)\}$

→ Instead we name $\{(e_i, e_j)\} \equiv V \otimes V$ with basis $e_i \otimes e_j$

$$\psi: V \times V \rightarrow V \otimes V$$

$$\varphi: (e_i, e_j) \mapsto e_i \otimes e_j$$

If $u, v \in V$ s.t.

$$u = \sum_i u_i e_i, \quad v = \sum_j v_j e_j$$

$$\rightarrow u \otimes v = \sum_{i,j} u_i v_j (e_i \otimes e_j)$$

$(V \otimes V, \varphi)$: satisfy a universal Property

$\rightarrow$ Any Bilinear, $B: V \times V \rightarrow K$ factorizes through the map φ giving rise to $\tilde{B}: V \otimes V \rightarrow K$.

$$B: V \times V \rightarrow W$$

$$\begin{array}{ccc} V \times V & \xrightarrow{\varphi} & V \otimes V \\ & \searrow B & \downarrow \tilde{B} \\ & & W \end{array}$$

$$\tilde{B} \circ \varphi = B$$

$$\bullet V^{\otimes 3} \equiv V \otimes V \otimes V$$

$$\bullet V^{\otimes} \equiv \bigoplus_{i=0}^{\infty} V^{\otimes i}, \quad V^{\otimes 0} \equiv K, \quad V^{\otimes 1} \equiv V$$

$$\ast \otimes: \otimes^{\ast} V \times \otimes^{\ast} V \rightarrow \otimes^{\ast} V$$

$$(e_{i_1} \otimes \dots \otimes e_{i_r}) \otimes (e_{j_1} \otimes \dots \otimes e_{j_s}) \mapsto (e_{i_1} \otimes \dots \otimes e_{i_r+s})$$

$B: V \otimes V \rightarrow K$, if it is symmetric, then it is because

B vanishes on the subspace spanned

by

$$\{e_i \otimes e_j - e_j \otimes e_i\}$$

$\rightarrow$ If it is anti symmetric it vanishes on the space spanned by $\{e_i \otimes e_i\}$

Proof:

$$= B(e_i \otimes e_j) = B(e_j \otimes e_i) = 0$$

$$\text{or } B(e_i \otimes e_j) = B(e_j \otimes e_i)$$

$$(ii) B(e_i \otimes e_i) = 0$$

$$\text{or } B(e_i \otimes e_i) = -B(e_i \otimes e_i)$$

$$\text{Sym}^2 V \equiv \frac{V \otimes V}{\text{Span}(\{e_i \otimes e_j - e_j \otimes e_i\})}$$

$$\Lambda^2 V \equiv V \otimes V / \text{Span}(\{e_i \otimes e_i\})$$

* A symmetric, bilinear map is an element of the vector space

$$(\text{Sym}^2 V)^*$$

$$A \text{ 2-form} \in (\Lambda^2 V)^*$$

$$\rightarrow \text{Sym}^\bullet V = \otimes^\bullet V / \text{gen}(\{e_i \otimes e_j - e_j \otimes e_i\})$$

$$\Lambda^\bullet V = \otimes^\bullet V / \text{gen}(\{e_i \otimes e_i\})$$

$\text{gen}(\{a_\alpha\})$ denotes the ideal of the algebra ~~generated~~ $\otimes^\bullet V$ generated by the elements $\{a_\alpha\}$

$$\text{where } \otimes^\bullet V = K \oplus V + V \otimes V + \dots$$

* Recall the ideal is a subspace

$$U \subseteq V, \text{ s.t. } u \otimes v \in U \quad \forall v \in V, u \in U$$

This means $\text{gen}(\{e_i \otimes e_i\})$

all elements $u \in \otimes^\bullet V$ s.t.

$$\{e_i \otimes e_j\} \otimes v \in \text{gen}\{I\}$$

• Consider $V^* : \wedge^0 V^*$ is formally defined as the Quotient of $\otimes^0 V^*$ by an ideal I

$$I = \text{reg}(\{e_i \otimes e_i\})$$

For $\alpha_1, \dots, \alpha_k \in V^*$, denote

$$\alpha_1 \wedge \dots \wedge \alpha_k \equiv \{\alpha_1 \otimes \dots \otimes \alpha_k\} \in \otimes^k V^* / I$$

Define $\text{Alt}_k : \wedge^k V^* \rightarrow \otimes^k V^*$

$$\text{Alt}_k(\alpha_1 \otimes \dots \otimes \alpha_k) := \sum_{\sigma \in \Sigma_k} \text{sgn}(\sigma) \alpha_{\sigma(1)} \otimes \dots \otimes \alpha_{\sigma(k)} \in \otimes^k V^*$$

Ex: σ : Permutation

$\alpha_1, \alpha_2 \in V^*$, then

$$\alpha_1 \wedge \alpha_2 \in \wedge^2 V^* \subset \otimes^2 V^* / I$$

$$\text{Alt}_2(\alpha_1, \alpha_2) = \alpha_1 \otimes \alpha_2 - \alpha_2 \otimes \alpha_1$$

→ Often we will simply ignore Alt_k notation and write

$$\alpha_1 \wedge \alpha_2 = \alpha_1 \otimes \alpha_2 - \alpha_2 \otimes \alpha_1$$

We can "lift" this map to $\wedge^0 V^*$

$$\text{Alt} \equiv \bigoplus_{n=0}^{\infty} \text{Alt}_n : \otimes^0 V^* \longrightarrow \otimes^0 V^*$$

Claim: $\text{Ker}(\text{Alt}) \supset I = \text{gen}(\{e_i \otimes e_i\})$

Proof: We may do this for the first few. To really prove this we would use induction:

⇒ Suppose $\alpha \in \text{Ker}(\text{Alt})$

then

$$\text{Alt}(\alpha) = \bigoplus_{n=0}^{\infty} \text{Alt}_n(\alpha) = 0$$

α can be a linear combination of forms and each will vanish at the respective level.

$$\sum \text{Sgn}(\sigma) \alpha_{\sigma_1} \otimes \dots \otimes \alpha_{\sigma_n} = 0$$

$$n=2: \alpha_1 \otimes \alpha_2 = \alpha_2 \otimes \alpha_1$$

$$\Leftrightarrow \alpha_1 = \alpha_2 \Leftrightarrow \alpha_1 \otimes \alpha_2 + \dots = \alpha$$

$$\therefore \alpha \in I$$

$n=3$: Exactly the same argument

$\Rightarrow$ Trivial to go the other way:

Assume $\alpha \in I$, then

$$\alpha = e_i \otimes e_i \otimes \dots$$

$$\Delta \text{Alt}(\alpha) = 0.$$

$$\alpha_1 \wedge \dots \wedge \alpha_n \mapsto \sum_{\sigma \in \Sigma_n} \text{Sgn}(\sigma) \alpha_{\sigma_1} \otimes \dots \otimes \alpha_{\sigma_n} \in \otimes^n V^*$$

Def: Interior Product / Contraction:

Let $X \in V$, (any),

$$i: V \times (\otimes^k V^*) \rightarrow \otimes^{k-1} V^*$$

$$i_X: \otimes^k V^* \rightarrow \otimes^{k-1} V^*$$

$$(\alpha_1 \otimes \dots \otimes \alpha_k) \mapsto \alpha_1(X) \alpha_2 \otimes \dots \otimes \alpha_k$$

On $\wedge^k V^*$:

$$i_X(\alpha_1 \wedge \dots \wedge \alpha_k) = \sum_{j=1}^k (-1)^j \alpha_j(X) \alpha_1 \wedge \dots \wedge \hat{\alpha}_j \wedge \dots \wedge \alpha_k$$

* Free Vector Space: K

Let S be a set. Then its free vec space

$V(S)$ has as a basis the set S ,

if α is a basis

→ Consider the set of maps

$$f: S \rightarrow K$$

which have finite support

ex: $|S| = n$, then

$$K(S) \cong K^n$$

ex: $|S| = \infty$, $|K(S)| = \infty$

* We can define the tensor product of spaces $V \otimes W$ as a quotient of $K(V \times W)$ by a subspace describing the bilinearity of interest.

$$Z \cong \text{span}(\{(v, w) + (v', w) - (v + v', w), (v, w) + (v, w') - (v, w + w'), (\lambda v, w) - \lambda(v, w), (v, \lambda w) - \lambda(v, w) \mid v \in V, w \in W, \lambda \in K\})$$

$$V \otimes W \equiv K(V \times W) / Z$$

$$(\otimes^2 V)^* \cong \otimes^2 V^*, (\text{Sym}^2 V)^* \cong \text{Sym}^2 V^*, (\wedge^2 V)^* \cong \wedge^2 V^*$$

ex: $V = \mathbb{R}^{2n}$, w/ canonical basis $\{e_i\}$ and dual $\{e^i\}$ has a canonical symplectic structure

$$\omega = \sum_{i=1}^n e^i \wedge e^{i+n}$$

$$\bullet \omega \in \text{End}(V) \Rightarrow \omega \in V \otimes V^*$$

$\bullet (V, J)$ and a basis (x_k, y_k) s.t. $y_k = Jx_k$
 $1 \leq k \leq \dim(V)/2$

$$\mathbb{R}^{2n} \rightarrow \mathbb{C}^n$$

$$z_k = x_k + i y_k$$

$$\bar{z}_k = x_k - i y_k = x_k + i (-y_k)$$

$$\text{Span}\{z_k, \bar{z}_k\} = V^{\mathbb{C}}$$

Linear transformations: